

MOHIT LINGARKAR

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Web: mowhits.github.io 

Github: [mowhits](https://github.com/mowhits) 

Education

- **Ramaiah Institute of Technology**, Bengaluru, India 2023 – Present
Bachelor of Engineering, Electronics and Communication Engineering CGPA: 9.50

Skills

- **Programming languages:** Python, C/C++, MATLAB, Verilog
- **Tools:** Linux, vim, bash, git, numpy, SPICE, Qiskit, Xilinx Vivado, Autodesk Fusion, $\text{\LaTeX}$, Inkscape

Awards and Achievements

- **Certificate of Distinction**, CeNSE Winter School on Semiconductor Technology 2024 Dec. 2024
- **Attendee**, QubitxQubit Summer School 2020 Nov. 2020

Projects

- **aes**  Nov. 2025
Designed AES encryption core in SkyWater SKY130 (130nm) node using open-source EDA tools such as Yosys, OpenROAD and LibreLane. Implemented pipelined architecture in SystemVerilog and achieved timing closure across all PVT corners and met all manufacturing checks (DRC/LVS clean), producing tapeout-ready GDSII spanning 2.7mm².
- **kilo-clone**  Apr. 2024
Developed a terminal-based text editor in C, replicating core functionalities of the 'kilo' editor. Implemented features such as file I/O, real-time screen rendering, and user input handling. Gained experience with low-level terminal control using VT100 escape sequences and dynamic memory management.
- **ams**  Dec. 2022
Designed a custom interpreted language in Python, implementing a lexer, parser combinators, and an AST to support control flow, expressions, and variable scope. Gained deeper insight into language design, functional parsing, and program evaluation.
- **recce**  Nov. 2021
Built an interactive music data analysis and clustering system using Python and Spotify data. Applied K-Means, t-SNE, and PCA to cluster and visualize tracks based on audio features. Integrated Spotify's API via `spotipy` for real-time song metadata retrieval. Created interactive visualizations with Plotly to explore musical trends across decades.